INTRODUCTION TO DATA SCIENCE (IDS)

Predicting Term Deposit Subscriptions Using Machine Learning in R

A Bank Marketing Analytics Approach

Project Review: Source Document / Content Blueprint for PPT Generation

Technology: R / R Studio | Dataset: UCI Bank Marketing

Team Members	[Names & Roll Numbers]
College	[College Name]
Department	[Department]
Academic Year	[20XX–XX]
Guide	[Faculty Name]

Instructions for the PPT generator

- Create a **12-slide** professional college project-review deck from Section 15 (slide-by-slide content). Other sections supply supporting detail.
- Theme: banking + data science. Navy (#0B2545) and teal (#13808B) on white, one accent colour, consistent icons. Max 4–6 short bullets per slide.
- Everything here is a **proposed methodology**. **Do NOT invent numbers.** Any value shown as _____ must stay a placeholder labelled “To be filled after model execution”.
- Technology is **R / R Studio**, not Python.
- Use diagrams for: pipeline flow, architecture, confusion matrix, preprocessing steps, package-to-stage map.

1. Project Understanding

A bank phones customers to sell a term deposit. Most calls fail. We use past campaign data, where we already know who said yes and who said no, to train a model. The model then predicts whether a new customer is likely to say yes, so the bank can focus on the right people.

Question	Answer
Term deposit	A deposit where the customer locks money with the bank for a fixed period at a fixed interest rate, with a penalty for early withdrawal. Gives the bank stable funds and the customer guaranteed returns.
Bank marketing campaign	An organised effort (here, mostly telemarketing calls) to contact customers and promote a product.
Bank's problem	Calling everyone costs agent time, money and customer goodwill, yet only a small fraction subscribe.
Why prediction helps	Rank customers by likelihood, call the best prospects first, skip low-probability ones.
What we predict	Whether a customer will subscribe: **yes** or **no**.
Target variable	y (term deposit subscription: yes/no).
Input features	Customer profile (age, job, marital, education, default, balance, housing, loan) and campaign data (contact, day, month, duration, campaign, pdays, previous, poutcome).
Role of ML	Finds patterns between features and outcomes in historical data and applies them to new customers.
Why R Studio	R is built for statistics and data analysis, with strong packages for wrangling (dplyr), visualisation (ggplot2) and modelling (caret, rpart, randomForest). R Studio provides scripts, console, plots and R Markdown reports in one IDE.
Who benefits	Bank marketing teams, call-centre managers, and customers (fewer irrelevant calls).
Real-world application	Lead scoring in CRM, call prioritisation, campaign budgeting, cross-selling other products.

2. Problem Definition

Banks run phone-based marketing campaigns to persuade customers to subscribe to term deposits. Contacting every customer is expensive, time-consuming and inefficient, because the subscription rate is low and repeated calls can annoy customers.

Formal definition: Given a customer's demographic, financial and campaign-related attributes, build a **binary classification model** in R that predicts whether the customer will subscribe to a term deposit (y = yes/no).

- **In scope:** analysis, preprocessing, EDA, model building, comparison and evaluation in R.
- **Out of scope:** real-time deployment, a live banking system, customer-level financial advice.

3. Project Objectives

1. Study the structure and quality of historical bank marketing data.

2. Clean and preprocess the data: unknown values, duplicates, outliers, encoding.

3. Use EDA to identify factors associated with term-deposit subscription.

4. Address class imbalance and split data into training and test sets.

5. Build classification models in R: Logistic Regression, Decision Tree, Random Forest, optionally KNN.

6. Evaluate and compare models using Confusion Matrix, Precision, Recall, F1-score and ROC-AUC, not accuracy alone.

7. Select the best model and translate its predictions into marketing recommendations.

4. Dataset Description

Source: UCI Machine Learning Repository, Bank Marketing dataset (Moro, Cortez & Rita, 2014), from direct phone marketing campaigns of a Portuguese banking institution.

Dataset versions (confirm which one you use)

File	Records	Input features	Notes
`bank-full.csv`	45,211	16	Original full dataset
`bank.csv`	4,521	16	10% sample of the above
`bank-additional-full.csv`	41,188	20	Adds economic indicators (employment rate, CPI, Euribor, etc.)
`bank-additional.csv`	4,119	20	10% sample of the above

Files are semicolon-separated: read.csv(..., sep = ";"). Confirm row count, column count and class ratio from your own file using dim() and table() before writing them on slides.

Feature table (assumes bank-full.csv: 16 inputs + target)

Feature	Type	Description
age	Numerical	Customer age
job	Categorical	Occupation type (admin., technician, blue-collar, student, retired, etc.)
marital	Categorical	Marital status
education	Categorical	Education level (primary, secondary, tertiary, unknown)
default	Categorical	Has credit in default? (yes/no)
balance	Numerical	Average yearly balance (euros)
housing	Categorical	Has housing loan?
loan	Categorical	Has personal loan?
contact	Categorical	Contact communication type (cellular, telephone, unknown)
day	Numerical	Day of month of last contact
month	Categorical	Month of last contact
duration	Numerical	Last contact duration in seconds
campaign	Numerical	Number of contacts during this campaign
pdays	Numerical	Days since last contact in a previous campaign (-1 = not previously contacted)
previous	Numerical	Number of contacts before this campaign
poutcome	Categorical	Outcome of previous campaign (success, failure, other, unknown)
y	**Target**	Subscribed to term deposit? (yes/no)

- **Customer profile:** age, job, marital, education
- **Financial status:** default, balance, housing, loan
- **Current campaign:** contact, day, month, duration, campaign
- **Previous campaign:** pdays, previous, poutcome
- **Numerical (7):** age, balance, day, duration, campaign, pdays, previous
- **Categorical inputs (9):** job, marital, education, default, housing, loan, contact, month, poutcome

Important note on `duration`

Call duration is only known after the call ends, so it cannot be used to decide whom to call. The dataset documentation says it should be dropped for a realistic predictive model. Either exclude it, or keep it and state clearly that the model is a benchmark. A good approach is to run both versions and compare. Faculty often ask about this.

Class imbalance

The “yes” class is a small minority (roughly 12% in the full version). Verify the exact figure with `prop.table(table(df$y))`.

5. Complete Methodology

5.1 Technical workflow, stage by stage

#	Stage	What happens	Why required	Done in R Studio	Output
1	Bank Marketing Dataset	Select UCI dataset, understand context	Defines problem and data	Read dataset documentation	Chosen dataset version
2	Data Collection	Download and import the CSV	Data must be in R	`read.csv(sep=";")`, `readr::read_delim()`	Data frame
3	Data Cleaning	Inspect structure; handle "unknown", duplicates, wrong types	Dirty data gives unreliable models	`str()`, `summary()`, `duplicated()`, `is.na()`, dplyr	Clean data frame
4	Data Preprocessing	Factors, outliers, encoding, scaling (KNN)	Algorithms need consistent input	`as.factor()`, `mutate()`, `scale()`, `dummyVars()`	Model-ready dataset
5	EDA	Visualise distributions and relationships with y	Reveals patterns, guides feature choice	ggplot2	Charts and insights
6	Feature Selection / Engineering	Drop leaky/useless variables (e.g. duration); create age groups, previously_contacted flag	Improves validity and interpretability	`mutate()`, `cut()`	Final feature set
7	Train-Test Split	Divide data (70/30 or 80/20), stratified by y	Test on unseen data	`caret::createDataPartition()`	train and test sets
8	ML Models	Train Logistic Regression, Decision Tree, Random Forest, (KNN)	Compare learning approaches	`glm()`, `rpart()`, `randomForest()`, `caret::train()`	Trained models
9	Prediction	Predict classes and probabilities on test data	Needed to compare with actuals	`predict()`	Predicted labels/ probabilities
10	Model Evaluation	Confusion matrix, Precision, Recall, F1, ROC-AUC	Quantify quality beyond accuracy	`confusionMatrix()`, `pROC::roc()`, `auc()`	Metrics table, ROC curves
11	Best Model Selection	Choose by agreed criteria (F1/AUC, interpretability)	One model needed for recommendations	Compare results table	Selected model
12	Subscription Prediction	Apply model to new customers	Practical output	`predict(model, newdata)`	Yes/No + probability
13	Business Recommendation	Convert predictions to campaign actions	Value comes from decisions	Interpret results, importance plots	Targeting strategy

5.2 Pipeline (ML Methodology slide)

Data → Preprocessing → EDA → Train/Test Split → Model Training → Prediction → Evaluation → Model Comparison → Final Prediction

This is **supervised learning**, specifically **binary classification**, because the labels are known and the target has two classes (yes/no).

5.3 How the Machine Learning Model Works (viva explanation)

Simple version (say this first)

"Our model is like an experienced bank employee who has seen thousands of past calls. From those calls it learned which kinds of customers usually say yes. When a new customer's details come in, such as age, job, education, balance, loans and previous campaign outcome, the model compares them with the patterns it learned and predicts **Yes (likely to subscribe)** or **No (unlikely to subscribe)**."

Technical version

- **Training data:** features (X) with known labels (y = yes/no).
- **Learning:** the algorithm searches for the relationship between X and y. Logistic Regression learns a weight per feature and converts the weighted sum into a probability with the sigmoid function. A Decision Tree learns if-else splits that best separate yes from no. Random Forest builds many trees on random samples/features and takes a majority vote.

- **Output:** for a new customer the model gives a **probability** of "yes". A **threshold** (default 0.5) converts it to a class. Lowering the threshold catches more likely subscribers (higher recall).
- **Testing:** check predictions on test data the model never saw, to judge generalisation.
- **Overfitting control:** train/test split, tree pruning (`cp` in rpart), cross-validation.

5.4 Technology Stack

Category	Tool	Purpose
Language	R	Statistical computing and ML
IDE	R Studio	Scripts, console, plots, R Markdown reports
Data import	`readr` / base R	Load the CSV
Data manipulation	`dplyr`, `tidyverse`	Filter, mutate, group, summarise, clean
Visualisation	`ggplot2`	Bar charts, histograms, boxplots, stacked bars
Correlation	`corrplot` (optional)	Correlation matrix of numeric variables
Splitting / training	`caret`	Partitioning, cross-validation, unified model interface
Models	`rpart`, `rpart.plot`, `randomForest`, base `glm`	Decision tree, tree plot, random forest, logistic regression
Imbalance (optional)	`ROSE` / `DMwR` / caret sampling	Over/under-sampling
Evaluation	`caret::confusionMatrix`, `pROC`	Confusion matrix, ROC curve, AUC

6. Data Preprocessing

Step	R approach	Purpose
Import	`read.csv("bank-full.csv", sep=";")`	Load data
Structure	`str()`, `dim()`, `head()`	Check types and size
Summary statistics	`summary()`	Detect odd ranges and distributions
Missing / "unknown"	`colSums(is.na(df))`; `table(df$job)`	Few true NAs, but "unknown" is used as a category. Options: keep as its own level, impute with mode, or drop rows. Document the choice and why.
Duplicates	`sum(duplicated(df))`; `distinct()`	Remove repeated records
Factor conversion	`mutate(across(where(is.character), as.factor))`	Models need categorical variables as factors
Outliers	`boxplot()`, IQR rule (balance, campaign, duration)	Extreme values can skew models. Cap or keep with justification; large balances may be genuine.
Encoding	`dummyVars()` / automatic factor handling	Needed for KNN and numeric-only algorithms
Scaling	`scale()` or `preProcess(method=c("center","scale"))`	Required for KNN (distance-based)
Class imbalance	Stratified split; optional up/down-sampling or ROSE on **training set only**	Prevents the model favouring "no"
Train-test split	`createDataPartition(df$y, p=0.7, list=FALSE)`	Fair evaluation on unseen data

Key rule: apply resampling to the training data only, never to the test set.

7. EDA Plan

Analysis	Visualisation	Insight sought
Target distribution (`y`)	Bar chart / pie	How imbalanced the data is
Age distribution	Histogram + density	Dominant age ranges; compare with subscribers
Job vs subscription	Stacked / proportional bar	Occupations with higher subscription rates

Analysis	Visualisation	Insight sought
Education vs subscription	Stacked bar	Does education level matter?
Balance vs subscription	Boxplot (log scale or trimmed)	Do subscribers hold higher balances?
Housing loan vs subscription	Proportional bar	Does a housing loan reduce interest?
Personal loan vs subscription	Proportional bar	Same question for personal loans
Contact method vs subscription	Bar chart	Cellular vs telephone effectiveness
Previous outcome (`poutcome`) vs subscription	Stacked bar	Do past successes predict future success?
Duration vs subscription	Boxplot	Longer calls correlate with yes, but this is a leakage warning
Month vs subscription	Bar chart	Seasonal patterns
Numeric correlations	Correlation heatmap	Redundant variables (e.g. pdays and previous)

Use **proportions** (`position="fill"`), not raw counts, when comparing categories, because raw counts are dominated by "no".

8. Machine Learning Algorithms

Model	Why consider it	Strength	Weakness
Logistic Regression	Standard baseline for binary classification	Simple, fast, interpretable coefficients and probabilities	Assumes linear relationship in log-odds
Decision Tree (rpart)	Easy to visualise and explain	Human-readable rules, handles mixed data	Can overfit, unstable
Random Forest	Ensemble of many trees	Usually higher accuracy, feature importance, less overfitting	Less interpretable, slower
KNN (optional)	Distance-based, no training phase	Intuitive	Needs scaling, slow on large data, weak with many categorical features

Start with the first three. Add KNN only if time allows, and say so on the slide.

9. Model Evaluation

Evaluation flow:

Actual Values + Predicted Values → Confusion Matrix → Evaluation Metrics

	Predicted No	Predicted Yes
Actual No	TN (True Negative)	FP (False Positive)
Actual Yes	FN (False Negative)	TP (True Positive)

Metric	Formula	Simple meaning
Accuracy	(TP+TN) / total	Overall share of correct predictions
Precision	TP / (TP+FP)	Of customers predicted "yes", how many truly subscribed (avoids wasted calls)
Recall (Sensitivity)	TP / (TP+FN)	Of all actual subscribers, how many we found (avoids missed opportunities)
F1-score	2·P·R / (P+R)	Balance between precision and recall
ROC-AUC	Area under ROC curve	Ability to separate yes from no across all thresholds (0.5 = random, 1 = perfect)

Why accuracy alone is not enough: if about 88% of customers say "no", a model that always predicts "no" gets about 88% accuracy but finds **zero** subscribers. Recall, F1 and AUC matter more here. (Verify the imbalance figure on your data.)

Model comparison table: To be filled after model execution

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	____	____	____	____	____
Decision Tree	____	____	____	____	____
Random Forest	____	____	____	____	____
KNN (optional)	____	____	____	____	____

Selecting the best model: compare F1 and AUC first, then Recall, since missing subscribers is costly. Also consider interpretability and training time.

10. System Architecture

Input Bank Customer Data → Data Preprocessing → EDA → Feature Preparation → Machine Learning Models → Model Evaluation → Best Model → Subscription Prediction → Bank Marketing Decision

Block	What to show inside it
Input Bank Customer Data	UCI Bank Marketing CSV, 16 features + target
Data Preprocessing	Unknown handling, duplicates, factors, outliers
EDA	ggplot2 charts, subscription patterns
Feature Preparation	Feature selection, encoding, scaling, train/test split
ML Models	Logistic Regression, Decision Tree, Random Forest (KNN)
Model Evaluation	Confusion matrix, Precision, Recall, F1, ROC-AUC
Best Model	Selected on agreed criteria
Subscription Prediction	Yes/No plus probability for each customer
Bank Marketing Decision	Call high-probability customers first, skip low-probability ones

11. Expected Results

Expected findings (hypotheses to test, not results)

- Customers with a previously successful campaign outcome are likely to show higher subscription rates.
- Occupation, age group and loan status may be associated with subscription.
- Contact method and month may influence conversion.
- Ensemble models like Random Forest often outperform a single tree or linear model, but this must be confirmed on your data.
- Class imbalance may give high accuracy but low recall for the “yes” class.

Actual results: To be filled after model execution

Best Model	_____
Accuracy	_____
Precision	_____
Recall	_____
F1 Score	_____
ROC-AUC	_____
Top 3 important features	_____

12. Business Impact

Technical output	Business benefit
Probability score per customer	Prioritise high-potential customers; reduce unnecessary calls

Technical output	Business benefit
Ranked customer list	Better customer targeting; improved campaign efficiency
Fewer wasted contacts	Lower campaign cost (agent hours, telecom)
Higher share of calls to likely subscribers	Higher conversion rate
Feature importance and EDA insights	Data-driven decisions on timing, channel, segment
Fewer repeated calls to unlikely customers	Better customer experience

13. Limitations

- Dataset is from one Portuguese bank and may not represent other banks or populations.
- Historical data may contain bias, such as who the bank chose to call.
- Customer behaviour changes over time, so models need retraining.
- Class imbalance can degrade performance on the minority “yes” class.
- Predictions are probabilistic, not guaranteed.
- Some variables, such as `duration`, are unavailable before the call, limiting real-time use.
- Data is from 2008–2010 and may not reflect current economic conditions.

14. Future Scope

- Hyperparameter tuning (grid/random search with caret)
- Advanced ensembles (gradient boosting, XGBoost)
- Better imbalance handling (SMOTE, cost-sensitive learning, threshold tuning)
- Feature engineering and selection
- Customer segmentation using clustering
- Real-time prediction
- Deployment as a web app (R Shiny)
- Integration with bank CRM systems

15. Complete 12-Slide PPT Content

For each slide: Title, Main Content, Key Points, Suggested Visual, Speaker Explanation.

Slide 1: Title Slide

Main Content	• Predicting Term Deposit Subscriptions Using Machine Learning in R: A Bank Marketing Analytics Approach • Subject: Introduction to Data Science (IDS) | Technology: R / R Studio • Team: [Names & Roll Nos.] | College: [College Name] | Department: [Department] | Academic Year: [20XX–XX] | Guide: [Faculty Name]
Key Points	Subject, technology, team, college details.
Suggested Visual	Navy banner with a bank/graph icon and the R logo.
Speaker Explanation	Good morning. Our project predicts which bank customers are likely to subscribe to a term deposit, using machine learning in R.

Slide 2: Introduction

Main Content	• Banks rely on marketing to sell deposit products. • Term deposit: money locked for a fixed period at a fixed rate. • Mass calling is costly, so targeted marketing is needed. • Data science finds patterns in past campaigns to guide future ones.
Key Points	Customer data + campaign history = predictive insight.
Suggested Visual	Icon row: Customer → Call → Deposit → Data.
Speaker Explanation	Banks collect a lot of data. If we learn from past campaigns, we can contact the right customers.

Slide 3: Problem Statement

Main Content	• Banks contact customers to promote term deposits. • Contacting everyone is expensive, time-consuming and inefficient. • **Goal:** build an ML model that predicts whether a customer will subscribe.
Key Points	Low conversion rate | Wasted effort | Need for data-driven targeting.
Suggested Visual	Funnel: many customers called → few subscribe.
Speaker Explanation	The problem is inefficiency. Our solution is a prediction model.

Slide 4: Objectives

Main Content	• Analyse historical bank marketing data. • Preprocess data and perform EDA. • Identify factors influencing subscription. • Build classification models in R. • Evaluate with metrics suited to imbalanced data. • Compare models and select the best. • Support efficient, data-driven marketing.
Key Points	Group as: Data → Model → Business.
Suggested Visual	Numbered icon list.
Speaker Explanation	We follow the full data science lifecycle, from raw data to a business recommendation.

Slide 5: Dataset Description

Main Content	• UCI Bank Marketing Dataset (Portuguese bank, phone campaigns). • Records: [confirm, e.g. 45,211 for bank-full]. Features: [confirm, 16 + target]. • Target: y (yes/no). • Include the feature table from Section 4.
Key Points	Customer profile | Financial status | Campaign data | Previous outcome | Class imbalance noted.
Suggested Visual	Feature table + small pie of y distribution (from your data).
Speaker Explanation	Each row is a customer contact. The target says whether they subscribed.

Slide 6: Data Preprocessing

Main Content	• Import (`read.csv`) → `str()` / `summary()` → handle "unknown" → remove duplicates → factors → outliers → encoding/scaling → imbalance handling → train/test split.
Key Points	dplyr, `caret::createDataPartition` | Resample training data only.
Suggested Visual	Horizontal step flow with R function labels.
Speaker Explanation	Raw data is rarely model-ready. These steps make it consistent and fair.

Slide 7: Exploratory Data Analysis

Main Content	• Table of the analyses and charts from Section 7 (age, job, education, balance, housing, loan, contact, poutcome, duration).
Key Points	Use proportions, not counts | Watch for leakage with `duration`.
Suggested Visual	2×3 grid of your actual ggplot2 charts (insert after running).
Speaker Explanation	EDA tells us which factors seem to matter before we build models.

Slide 8: Machine Learning Methodology

Main Content	• Pipeline: Data → Preprocessing → EDA → Split → Train → Predict → Evaluate → Compare → Final Prediction. • Binary classification: Yes / No. • Models: Logistic Regression, Decision Tree, Random Forest, (KNN).
Key Points	Supervised learning on labelled history.
Suggested Visual	Horizontal pipeline flowchart (9 blocks).
Speaker Explanation	This is supervised learning. The model learns from labelled history.

Slide 9: Model Building in R Studio

Main Content	• Packages: tidyverse, dplyr, ggplot2, caret, rpart, rpart.plot, randomForest, pROC. • Process: train on training set → tune using cross-validation → predict on test set.
Key Points	No code on slide; explain what training does.
Suggested Visual	Package icons mapped to pipeline stages.
Speaker Explanation	Each package has a specific role in our workflow.

Slide 10: Model Evaluation

Main Content	• Confusion matrix diagram, metric definitions, and why accuracy alone is misleading. • Flow: Actual + Predicted → Confusion Matrix → Metrics.
Key Points	Accuracy, Precision, Recall, F1, ROC-AUC.
Suggested Visual	2×2 confusion matrix + empty comparison table ("To be filled after model execution").
Speaker Explanation	For imbalanced data, recall, F1 and AUC tell us more than accuracy.

Slide 11: Expected Results / Findings

Main Content	• Questions we will answer: which characteristics matter, which factors influence conversion, which model is best, which customers to target. • Results box: Best Model _____ | Accuracy _____ | Precision _____ | Recall _____ | F1 _____ | AUC _____.
Key Points	Hypotheses, not results, until the code is run.
Suggested Visual	Placeholder results card + feature-importance chart (after execution).
Speaker Explanation	These are the outcomes we are measuring. [If results are ready, state them here.]

Slide 12: Conclusion & Future Scope

Main Content	• Conclusion: ML in R can predict subscriptions and support data-driven banking decisions. • Future scope: tuning, boosting, SMOTE, feature engineering, segmentation, Shiny app, CRM integration.
Key Points	Achieved vs next steps.
Suggested Visual	Two columns: “Achieved” and “Next steps” + Thank You.
Speaker Explanation	We showed an end-to-end approach. The next step is deployment and refinement.

16. Detailed Speaker Notes (flow guidance)

About 1 minute per slide, 12–14 minutes total.

1. **Title:** Introduce the team and topic in one sentence.
2. **Introduction:** Explain a term deposit in simple words (a fixed deposit). Link it to marketing cost.
3. **Problem:** Emphasise “contacting everyone is inefficient.” State the prediction goal clearly.
4. **Objectives:** Do not read each one. Group them as data, model and business.
5. **Dataset:** Mention source and target. State honestly which version you used and its size. Mention `duration` leakage to show awareness.
6. **Preprocessing:** Highlight “unknown” values and imbalance as the two interesting issues.
7. **EDA:** Pick 2–3 charts and state one insight each from your real charts.
8. **Methodology:** Walk the pipeline left to right. Explain supervised binary classification.
9. **Model building:** Don’t show code. Explain what happens in training.
10. **Evaluation:** Use the “always predict no” example to justify recall and F1.
11. **Results:** Present only real numbers. If not run yet, say these are expected outcomes.
12. **Conclusion:** Summarise in one sentence, then give 2–3 future-scope items.

17. Viva Questions & Answers

Q1. Why did you choose this problem?

It is a realistic business problem with a clear binary target, available public data, and direct value in reducing marketing cost.

Q2. Why is this a classification problem?

The target `y` has discrete categories (yes/no), not a continuous number.

Q3. What is the target variable?

`y`, whether the customer subscribed to a term deposit.

Q4. Why did you use R Studio?

R is designed for statistics and data analysis, with strong visualisation and modelling packages, and R Studio gives an integrated workspace.

Q5. Why not Python?

Python is also suitable. We used R because our course emphasises it and it offers concise, strong tools for statistical modelling and visualisation.

Q6. What dataset are you using?

The UCI Bank Marketing dataset from a Portuguese bank's phone campaigns. (State your version and size.)

Q7. What is EDA?

Exploratory Data Analysis: summarising and visualising data to understand patterns, anomalies and relationships before modelling.

Q8. Why is preprocessing required?

Real data has unknown values, duplicates, wrong types and outliers. Models need clean, consistent input.

Q9. What is class imbalance?

One class has far more samples than the other. Here "no" greatly outnumbers "yes", which can bias models toward "no".

Q10. What is logistic regression?

A model that estimates the probability of a class using a weighted sum of features passed through a sigmoid function.

Q11. Why use a decision tree?

It is easy to interpret: it produces if-else rules a bank manager can understand.

Q12. Why use random forest?

It combines many trees, which reduces overfitting and usually improves performance and stability.

Q13. What is overfitting?

When a model memorises training data, including noise, and performs poorly on new data. We reduce it with train/test split, cross-validation and pruning.

Q14. What is a confusion matrix?

A 2×2 table comparing actual and predicted classes (TP, TN, FP, FN).

Q15. What is precision?

Of the customers predicted "yes", the fraction that truly subscribed.

Q16. What is recall?

Of all actual subscribers, the fraction the model identified.

Q17. What is F1-score?

The harmonic mean of precision and recall, giving a single balanced measure.

Q18. What is ROC-AUC?

The ROC curve plots true-positive rate against false-positive rate across thresholds. AUC measures overall separating ability (0.5 = random, 1 = perfect).

Q19. How do you select the best model?
Compare F1, recall and AUC on the test set, then consider interpretability and complexity.

Q20. How can the bank use your prediction?
Rank customers by subscription probability, call the top segment first, and reduce calls to low-probability customers.

Q21. What are the limitations?
Single bank, older data, possible bias, class imbalance, probabilistic predictions, and some variables unavailable before the call.

Q22. What is the future scope?
Hyperparameter tuning, boosting models, SMOTE, segmentation, Shiny deployment and CRM integration.

Q23. What is the problem with the `duration` variable?
It is known only after the call, so using it leaks future information. A realistic model should exclude it.

Q24. Why split data into training and testing sets?
To evaluate on unseen data, so we measure generalisation, not memorisation.

Q25. What is stratified splitting?
Splitting while preserving the same class ratio in both sets, important for imbalanced data.

Q26. How did you handle "unknown" values?
(State your choice: kept as a category, imputed, or removed, and why.)

Q27. What is cross-validation?
Training and validating on multiple data folds to get a more reliable performance estimate.

Q28. What does `pdays = -1` mean?
The customer was not contacted in a previous campaign.

Q29. Why scale data for KNN?
KNN uses distances, so large-valued features such as `balance` would dominate if unscaled.

Q30. Can the model guarantee a customer will subscribe?
No. It gives probabilities, not certainties.

18. Recommended PPT Visuals / Diagrams

Slide	Visual
1	Navy banner with bank and R icons
2	Icon strip: Customer → Call → Deposit → Data
3	Funnel: contacted vs subscribed
4	Numbered icon list
5	Feature table + y distribution pie (from your data)
6	Horizontal preprocessing flow with R function tags
7	Grid of real ggplot2 charts: age histogram, job stacked bar, balance boxplot, poutcome bar
8	Pipeline flowchart (9 blocks)
9	Package-to-stage mapping diagram
10	2×2 confusion matrix + metric formula cards
10/11	ROC curve overlay for all models (after execution)
11	Model comparison table + feature-importance bar chart (after execution)
12	"Achieved vs Future" two-column layout
Architecture	Block diagram from Section 10

Use the same colour palette in ggplot2 (scale_fill_manual) as in the slides.

19. Final Project Workflow

Bank Marketing Dataset → Data Collection → Data Cleaning → Data Preprocessing → EDA → Feature Selection/Engineering → Train-Test Split → ML Models → Prediction → Model Evaluation → Best Model Selection → Term Deposit Subscription Prediction → Business Recommendation

R skeleton (reference for your script, not for slides)

```
library(tidyverse); library(caret); library(rpart)
library(rpart.plot); library(randomForest); library(pROC)

df <- read.csv("bank-full.csv", sep = ";", stringsAsFactors = TRUE)
str(df); summary(df); table(df$y)

set.seed(123)
idx   <- createDataPartition(df$y, p = 0.7, list = FALSE)
train <- df[idx, ]; test <- df[-idx, ]

glm_m <- glm(y ~ ., data = train, family = binomial)
tree  <- rpart(y ~ ., data = train, method = "class")
rf    <- randomForest(y ~ ., data = train)

pred <- predict(rf, test)
confusionMatrix(pred, test$y, positive = "yes")
```

Before building the PPT: run the code and confirm the dataset version, row count, class ratio and your `duration` decision, then fill in the placeholders with real values.